

Practice Quiz: Cognition

Part A: Multiple Choice

1. The C-vector encodes: A) Transition probabilities between states B) Log-preferences over observations C) Prior beliefs over initial states D) The agent's learning rate
2. If $\mathbf{C} = [0, 0, 0]$, the agent's behavior is driven entirely by: A) Risk (pragmatic value) B) Ambiguity (epistemic value) C) Habits (E-vector) D) Nothing — the agent is frozen
3. The D-vector is used in Active Inference as: A) The posterior after the first observation B) The initial prior for state inference C) The reward signal D) The transition dynamics prior
4. What role does the E-vector play in the policy posterior equation? A) It replaces EFE entirely B) It adds a log-prior bias to each policy's score C) It scales the precision parameter D) It modifies the A-matrix
5. When precision $\gamma = 0.01$, the agent's policy selection is: A) Nearly deterministic B) Nearly random (or dominated by habits if E is set) C) Identical to maximum EFE selection D) Always action 0
6. The risk component of EFE is $D_{\text{KL}}[q(o|\pi) \mid \tilde{P}(o)]$. The target distribution $\tilde{P}(o)$ is: A) The empirical observation frequency B) $\sigma(\mathbf{C})$ — the softmax of the C-vector C) The A-matrix likelihood D) A uniform distribution
7. `plot_D_prior()` annotates the bar chart with: A) The VFE value B) The entropy of D C) The number of states D) The D-vector norm

Part B: Short Answer

1. An agent has $\mathbf{C} = [0, 5, -5]$ for (neutral, food, predator). Its current beliefs are $q(s) = [0.5, 0.5]$. The A-matrix maps state 0 to observation 1 (food) and state 1 to

observation 2 (predator). Describe qualitatively what policy the agent should prefer and why, in terms of the risk component.

2. Write code to create an `ActiveInferenceAgent` with a habit prior that strongly favors policy 1 (go-left) and a low precision $\gamma = 0.1$. Explain why this agent would almost always go left regardless of observations.
3. Design an experiment to determine the "critical γ " at which an agent transitions from exploratory to exploitative behavior. Describe the setup, the measurement, and what plot you would produce.